Guided Notes — Ch. 4

Consumption, Saving, and Investment

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Fill in the blanks as we work through the slides; slide numbers in each heading refer to the lecture deck. Equations are given in full — you supply the reasoning. The italic question after each topic has no single right answer; jot a line and bring it to class.

Consumption and Saving

LECTURE 1 *§4.1 determinants of consumption & saving — slides 5–17*

The importance of consumption and saving *(slide 5)*

• Desired national saving: $S^d = Y - C^d - G$ (4.1).
• Saving is the part of income *not* consumed.
• Measured when C is at its desired level.
• National saving also includes government saving.

Can a country of thrifty households still have low national *saving? What would drive that?* [yes — a large government deficit (G high) can offset private thrift]

The consumption–saving decision of an individual *(slide 6)*

• Consuming less than income means saving is positive.
• Consuming more than income means saving is negative (dissaving).
• The two choices are linked because $S = Y - C$.

If consuming and saving are one decision, why model "the saving decision" on its own? [S is the residual; isolating it makes intertemporal / interest-rate analysis cleaner]

Current versus future consumption *(slide 7)*

• One unit of consumption today costs $1 + r$ units tomorrow.
• Saving shifts consumption from the present to the future.
• The consumption-smoothing motive favours an even path over time.

Why does a higher r make current *consumption "more expensive," and how does smoothing temper it?* [price of today's consumption is $1 + r$; smoothing means people won't swing fully to the future]

Effect of changes in current income *(slide 8)*

• When Y rises, C^d rises but by less than Y, so S^d rises.
• MPC (fraction of extra income consumed) lies between 0 and 1.
• Higher *expected future* income raises C today, so saving falls.

A permanent raise vs. a one-off bonus of equal size — which lifts current saving more, and why? [the one-off (temporary) income — smoothing spreads it, so more is saved now]

The Keynesian consumption function *(slide 9)*

• Consumption function: $C^d = c_0 + c_y Y$.
• $c_0 =$ autonomous consumption (C when $Y = 0$).
• $c_y =$ the MPC and the slope, with $0 < c_y < 1$.

A constant MPC misrepresents whose behaviour — the rich or the poor — and which way? [overstates the rich household's MPC, since MPC tends to fall as income rises]

Application: the pandemic recession (Fig. 4.1) *(slide 10)*

• Sentiment indexes give little warning before recessions.
• 2020 was the shortest recession in US history, and the deepest since the Great Depression.
• Governments replaced lost income, so income rose and saving rose.

If income rose *in 2020, in what sense was it still the "deepest" recession?* [output and employment collapsed; income was propped up by transfers, not by production]

Application: income, consumption, and saving (Fig. 4.2) *(slide 11)*

• Spending and confidence are correlated.
• But confidence does not improve forecasts from real-time data.
• In 2020, spending shifted from services to goods.

If confidence tracks spending but adds no forecasting power, is it useless to a policymaker? [no — still a fast real-time signal / narrative; it just adds no *marginal* predictive gain]

Effect of changes in wealth *(slide 12)*

• Higher wealth raises consumption and lowers current saving.
• Future resources are still higher despite the lower saving.
• The MPC out of wealth is small.

Why should the MPC out of wealth *be smaller than the MPC out of current income?* [wealth is spread over the whole lifetime (permanent-income logic), so each year's extra spending is small]

Effect of changes in the real interest rate *(slide 13)*

• Substitution effect: higher $r \Rightarrow$ more saving.
• Income effect, saver: higher $r \Rightarrow$ less saving.
• Income effect, borrower: higher $r \Rightarrow$ more saving.
• Net effect: theoretically ambiguous; empirically a small rise in saving.

For whom could a higher r lower *saving — and why is that not a contradiction?* [a saver, via the income effect; the substitution effect roughly offsets it]

Taxes and the real return to saving *(slide 14)*

• After-tax real rate: $r_{a\text{-}t} = (1 - t)\, i - \pi^e$ (4.2).
• A higher tax rate t lowers the return.
• Higher expected inflation π^e also lowers it.

Two savers, same nominal rate but different brackets and inflation — who saves more keenly? [the one with lower t and lower π^e, i.e. the higher after-tax real rate]

Calculating after-tax interest rates (Table 4.1) *(slide 15)*

• With $i = 5\%,\ \pi^e = 2\%$: at $t = 30\%$, $r_{a\text{-}t} =$ 1.5%.
• At $t = 20\%$, $r_{a\text{-}t} =$ 2%.
• Taxing nominal interest over-taxes savers under inflation.

Can the after-tax real return be negative *with a positive nominal rate? What then?* [yes, under high inflation + tax on nominal interest; it discourages saving, rewards borrowing]

In touch with data: interest rates *(slide 16)*

• Rates differ by default risk.
• They differ by term — the yield curve — and by tax status.
• Because rates move together, we say "the" interest rate.

If all rates move together, why distinguish default risk, term, and tax status at all? [the spreads carry information (risk, expectations, tax); "the" rate is only a simplification]

The yield curve (Fig. 4.A) *(slide 17)*

• The yield curve links a bond's term to the rate it pays.
• A normal curve slopes up.
• An inverted curve often signals a recession.

Why might an inverted curve help cause *a slowdown, not just predict one?* [it signals expected rate cuts / pessimism, which can dampen borrowing and spending]

LECTURE 2 *§4.1 fiscal policy (18–22) → §4.2 investment setup (24–31)*

Fiscal policy *(slide 18)*

• Affects C^d through current and expected future income.
• Affects national saving directly: $S^d = Y - C^d - G$.

Could the direct and indirect channels ever push desired national saving in opposite *directions?* [possibly — but a temporary $G \uparrow$ lowers both C^d and S^d, so the direct effect dominates]

Government purchases (temporary increase) *(slide 19)*

• Temporary $G \uparrow$ lowers after-tax income, so C^d falls.
• C^d falls by less than the rise in G.
• So S^d falls — $G \uparrow$ reduces both C^d and S^d.

If a temporary G rise lowers both C and S, who ultimately pays for the extra purchases? [current consumers (lower C) and future generations (lower investment ⇒ smaller capital stock)]

Taxes *(slide 20)*

• A tax cut today, paid by higher future taxes, raises current income.
• Future income falls, offsetting the current gain.
• So desired consumption may rise or fall.

Why is the timing *of taxes supposed to be irrelevant to a forward-looking household?* [only the present value of taxes matters, and a cut-now/tax-later swap leaves it unchanged]

Ricardian equivalence *(slide 21)*

• If the offset is exact, consumption is unchanged.
• A tax change alters only the timing of taxes (same present value).
• If people don't foresee future taxes, equivalence fails.
• Parker et al.: households spent 50–90% of the 2008–09 rebates.

Which groups break the "people foresee future taxes" assumption most, and how does the evidence bear on it? [credit-constrained / myopic households; they spent most of the rebate, contradicting equivalence]

Summary 5: determinants of desired national saving *(slide 22)*

• S^d *rises* when current income rises or the after-tax rate rises.
• S^d *falls* when future income, wealth, or G rises.
• The relevant rate is the after-tax real rate.

Which determinant moves C and S the same *way, and which move them oppositely — and why?* [current income ⇒ both rise; future income, wealth, G raise C but cut S]

Investment

Why is investment important? *(slide 24)*

Investment fluctuates sharply over the business cycle — it is the most volatile component of GDP — so we cannot understand the cycle without understanding investment. It also matters for the long run, because investment adds to the capital stock and so drives economic growth.

Investment is a small share of GDP yet dominates cyclical swings — how can something so small drive the cycle? [its swings are proportionally huge, so it contributes most of the *change* in GDP]

The desired capital stock *(slide 25)*

The *desired capital stock* is the amount of capital that lets a firm earn the largest expected profit. Because new investment becomes productive capital only with a lag, the benefit of investing is the *future* marginal product of capital, MPK^f, not today's.

Why is the future *MPK the right benefit of investing, and what does that put centre-stage?* [capital pays off later, with a lag — so expectations (and their uncertainty) drive investment]

The user cost of capital *(slide 26)*

Holding a unit of capital for a period has a real cost — the interest forgone plus wear and tear — called the *user cost*:

$$uc = rp_K + dp_K = (r + d)\, p_K \qquad (4.3),$$

that is, interest cost plus depreciation. For Kyle's Bakery, with $r = 8\%$, $d = 10\%$ and $p_K = 100$, the user cost is $uc =$ 18 per unit per year.

Could a firm face a high user cost even when interest rates are zero? [yes — high depreciation d or a high capital price p_K keeps uc up]

Determination of the desired capital stock (Fig. 4.3) *(slide 27)*

A firm keeps adding capital until the benefit just equals the cost, i.e. where $MPK^f = uc$. The MPK^f schedule falls as K rises (diminishing marginal productivity), while the user cost does not depend on K and so is flat (horizontal). If $MPK^f > uc$ the firm should add capital; if $MPK^f < uc$ it should shed capital.

Why can't a firm profitably keep adding capital past $MPK^f = uc$ even while MPK is still positive? [beyond that point $MPK^f < uc$: the last unit costs more than it earns, cutting profit]

Factors that change the desired capital stock *(slide 28)*

The desired capital stock changes whenever the MPK^f curve shifts or the user cost changes. The drivers are therefore the real interest rate, the depreciation rate, the price of capital p_K, and technological change that alters the future marginal product.

A rise in p_K and a rise in r both raise the user cost — do they act through the same channel? [both raise uc; but technology shifts the MPK^f curve itself, a different channel]

A decline in the real interest rate raises desired K (Fig. 4.4) *(slide 29)*

A fall in r lowers the user cost, shifting the uc line down, so the desired capital stock rises. For Kyle's Bakery, cutting r from 8% to 6% lowers uc from \$18 to \$16, and desired capital rises from 5000 to 6000 cubic feet.

If lower rates raise desired K, why did near-zero rates after 2008 fail to spark an investment boom? [weak demand and pessimistic expectations lowered MPK^f, offsetting the cheaper uc]

An increase in expected MPK^f raises desired K (Fig. 4.5) *(slide 30)*

A technological advance raises the expected future MPK^f, shifting that curve up, so desired capital rises (again 5000 to 6000). Not every new technology has this effect, however — some is capital-saving and can actually lower the desired capital stock.

Could a genuine technological breakthrough reduce *a firm's desired capital? When?* [yes — capital-saving technology lets the firm produce the same output with less capital]

Taxes and the desired capital stock *(slide 31)*

When output is taxed, the return the firm keeps is only $(1-\tau)MPK^f$. Setting this equal to the user cost gives the tax-adjusted rule

$$MPK^f = \frac{uc}{1-\tau} = \frac{(r+d)p_K}{1-\tau} \qquad (4.4).$$

A higher tax rate τ therefore raises the tax-adjusted user cost and reduces the desired capital stock.

Through what single channel does a higher corporate tax rate reduce investment here? [it raises the tax-adjusted user cost, $uc/(1-\tau)$]

LECTURE 3 *§4.2 investment, rest (32–39) → §4.3 goods market (41–45)*

Taxes and the desired capital stock (continued) *(slide 32)*

Because in reality it is profits, not revenues, that are taxed, depreciation allowances lower the effective burden, and investment tax credits reduce the tax on new investment. The whole tax code is summarised by the effective tax rate, and it is this effective, not the statutory, rate we should compare across countries. Cummins, Hubbard, and Hassett estimate an investment-to-user-cost elasticity of about -0.66.

Why can two countries with the same statutory rate have very different effective rates on capital? [depreciation allowances and investment tax credits differ, changing the effective burden]

From the desired capital stock to investment *(slide 33)*

The capital stock changes through two opposing channels: gross investment adds to it, while depreciation subtracts from it. The net change is what firms ultimately care about.

Under what condition can gross investment be positive while the capital stock still shrinks? [when depreciation exceeds gross investment (net investment < 0)]

Net investment *(slide 34)*

Net investment is gross investment minus depreciation,

$$K_{t+1} - K_t = I_t - dK_t \qquad (4.5),$$

and it equals the change in the capital stock. So gross investment can be positive while *net* investment is negative, whenever depreciation exceeds it.

Why does a stretch of negative net investment matter for future living standards? [a shrinking capital stock lowers future potential output]

Gross and net investment, 1929–2021 (Fig. 4.6) *(slide 35)*

The figure plots both series as a % of GDP. In the deepest downturns — parts of the Great Depression and World War II — net investment was actually negative, meaning the capital stock was shrinking, which lowers future output.

What does a period of negative net investment reveal about how deep those shocks were? [the economy was consuming its own capital faster than it replaced it]

From the desired capital stock to investment (continued) *(slide 36)*

Rearranging (4.5) gives $I_t = K_{t+1} - K_t + dK_t$. If firms could reach their target in a single period, then $K^* = K_{t+1}$, and so

$$I_t = K^* - K_t + dK_t \qquad (4.6).$$

Equation (4.6) assumes firms hit K^* in one period — why is that unrealistic, and what does slide 38 add? [installation lags; adjustment toward K^* is spread over several years]

Investment has two parts *(slide 37)*

Equation (4.6) splits investment neatly. The first part is the desired net increase in the capital stock, $K^* - K_t$; the second is the replacement of capital that has depreciated, dK_t. For a mature firm whose capital stock is near its target, replacement investment dominates.

For a fast-growing start-up, which of the two parts dominates, and why? [the net increase $K^* - K_t$, since its capital is far below target]

Lags and investment *(slide 38)*

Some capital can be installed quickly, but other capital takes years to put in place, so investment toward the desired stock is spread over several years. These lags delay and weaken investment's response to interest rates. (A related market signal is Tobin's q — the ratio of a firm's market value to the replacement cost of its capital; when $q > 1$, firms should invest more.)

If investment responds only slowly to r, what does that imply for monetary policy's power over investment? [its effect is delayed and uncertain — weaker and lagged than the textbook rule suggests]

Summary 6: determinants of desired investment *(slide 39)*

Desired investment *rises* when the future marginal product MPK^f rises, when the real interest rate or user cost falls, and when the effective tax rate falls. In short, investment is tied directly to the desired capital stock.

All three determinants of investment work through one common variable — what is it? [the desired capital stock (they all move MPK^f relative to uc)]

Goods Market Equilibrium

The goods market equilibrium condition *(slide 41)*

Goods-market equilibrium requires output to equal total demand,

$$Y = C^d + I^d + G \qquad (4.7).$$

Unlike the income–expenditure identity, this condition need not hold at every moment — undesired output can pile up as unplanned inventories. The real interest rate adjusts to clear the market. Equivalently, equilibrium is where desired saving equals desired investment, $S^d = I^d$ (4.8).

Why can the equilibrium condition fail while the income–expenditure identity *always holds?* [the identity counts unplanned inventories as investment; equilibrium needs desired = actual]

Plotting S^d vs. I^d (Fig. 4.8) *(slide 42)*

On the saving–investment diagram, equilibrium is where $S^d = I^d$. The saving curve slopes up in the real interest rate, while the investment curve slopes down. If r is too low, desired investment exceeds desired saving, and competition among borrowers for scarce funds pushes r up.

At a disequilibrium r, what concrete market force nudges r back toward equilibrium? [excess I makes borrowers bid r up; excess S makes lenders compete r down]

Components of aggregate demand: an example *(slide 43)*

In the numerical example, at $r = 6\%$ desired saving and investment are equal, $S^d = I^d = 1000$, and output equals demand, $Y = C^d + I^d + G = 4500$ — this is equilibrium. At $r = 3\%$, aggregate demand (5150) exceeds output (4500) and desired investment (1500) exceeds desired saving (850); raising r back to 6% restores equilibrium by cutting C and I and raising saving.

At $r = 3\%$, "demand > supply" and "$I^d > S^d$" both hold — are these the same statement? [yes — they are equivalent, since $Y = C + I + G$ rearranges to $S = I$]

A decline in desired saving (Fig. 4.9) *(slide 44)*

A fall in desired national saving — for example, a temporary G increase — shifts the saving curve left. The real interest rate rises (6% to 7%) and investment is crowded out (E to F). This crowding out need not be complete; how much investment falls depends on the slopes of the two curves.

Why is crowding out usually partial *rather than complete?* [the higher r also trims C and lifts S, so I falls by less than G rises]

An increase in desired investment (Fig. 4.10) *(slide 45)*

A rise in desired investment — say, an invention that raises the expected future MPK^f — shifts the investment curve right. The real interest rate rises (6% to 8%), and both saving and investment rise (E to G). At the higher rate, savers gain while borrowers pay more.

A higher-investment economy raises r (helping savers, hurting borrowers) — is it better off on net? [usually yes for the future (more capital ⇒ more output), but with a distributional trade-off today]

Application: the boom and bust in stock prices *(slide 57)*

Sharp swings in stock prices feed the goods market through two channels we have already met: a wealth effect on consumption (higher wealth raises C, lowers S) and a Tobin's-q effect on investment. A boom raises both; a bust reverses them. Whether central banks should lean against asset-price booms remains debated ("lean vs. clean").

When a bubble bursts, which channel bites faster and which is larger — consumption or investment? [the consumption/wealth channel acts fast; the q/investment channel lags but can be large]

Chapter Summary

Consumption and Saving

• Saving $= Y - C$, so consuming and saving are one decision; households save to move consumption to the future.
• ↑ current income raises both consumption and saving; ↑ future income or wealth raises consumption but lowers saving.
• Higher r: the substitution effect raises saving; the income effect raises it for borrowers, lowers it for savers — small net rise. Relevant rate $=$ after-tax real.
• Temporary ↑ G lowers C by less than ΔG, so national saving falls.
• Ricardian equivalence: a lump-sum tax cut leaves C, S unchanged, unless people ignore future taxes.

Investment and Goods Market Equilibrium

- The desired capital stock maximises profit, where $MPK^f = uc$, with $uc = (r + d)p_K$.
- A lower user cost, higher expected MPK^f, or lower effective tax rate raise desired K.
- Net investment = gross − depreciation $= \Delta K$; firms invest to reach the desired K.
- Equilibrium: $Y = C^d + I^d + G$, i.e. $S^d = I^d$; the real interest rate clears it.
- Saving–investment diagram: S slopes up, I slopes down; curve shifts change r.

Voices on Consumption, Saving & Investment

"Consumption—to repeat the obvious—is the sole end and object of all economic activity."
— J. M. Keynes, *The General Theory of Employment, Interest and Money* (1936), Ch. 8

"Most, probably, of our decisions to do something positive … can only be taken as a result of animal spirits."
— J. M. Keynes, *The General Theory* (1936), Ch. 12 — on investment

"The engine which drives Enterprise is not Thrift, but Profit."
— J. M. Keynes, *A Treatise on Money* (1930)

"Capitals are increased by parsimony, and diminished by prodigality and misconduct."
— Adam Smith, *An Inquiry into the Nature and Causes of the Wealth of Nations* (1776), Book II, Ch. III

"[The rate of interest is] determined by impatience to spend income and opportunity to invest it."
— Irving Fisher, *The Theory of Interest* (1930)